

Messaging around GeoPIXE modules using the "Notify" mechanism

Direction: "Incoming" for Notify messages that are handled here, "Outgoing" for Notify message sent from here, "Pass on" for message that are not handled, but are allowed to be passed on (Note 1)
Tag: The name of the Notify message

- Note 1 If a routine has Registered for some Notify message tags, then they get passed on automatically, even if the routine never processes them.
Note 2 Notify name given by argument when 'spectrum_display' opened, depends on how window used: "spectrum-display", "chart-display", "groups-display" (Maia Mapper).
Note 3 Notify name given by argument when 'detector_select' opened. Passed from <update_display> in 'spectrum_select'.

Program	Routine	Direction	Tag	Function	Notes
evt	evt_event	Incoming		Open the raw data sorting to images window	
			detector-fwhm	set channel enables based on channel FWHM below some threshold	Requires the detector layout table to include resolution values. Not really used anymore
	many	Outgoing	detector-load	detector layout loaded elsewhere - do you want to set channel enable based on new details?	e.g. from spectrum display/select no used anymore
			detector-select	switches display of detector channel parameters to a selected channel	
			device-update	update EVT window to reflect new device object parameters for detector	
			dpath	set a specific path for raw data files, which can be eslewhere (e.g. Maia blog files)	experimental, not used now
			path	set the default path for most data files, especially output files	
			snapshot	was to snapshot the position and size of a window	
			start-evt	Sent from Batch_Sort to start evt file processing	These contain a "command" to be executed and arguments returned to Batch_Sort
			wizard-action	a command from a Wizard using wizard mechanism (linked list of commands)	
			abort-evt	indicates "start-evt" has been aborted, which is returned to Batch_Sort	
			detector-get	"Get" has been used to load new Energy Cals	e.g. picked up Batch_Sort to adopt this new device returned to Batch_Sort
			device	device object has changed	
			done-evt	indicates "start-evt" has completed	
			dpath	indicates that the specific path for raw data files (e.g. Maia blog files) has changed	triggers a 'detector-load' in 'detector_layout'
			evt-load	a new DAI template has been loaded, anyone interested?	
			evt-mode	mode of processing (e.g. DA, CUTs, MPDA, ...) has changed	
			evt-type	type of data (e.g. SXRF, PIXE, ...) has changed	e.g. main GeoPIXE Image routine and clones
			images	a new image has been formed and pointer passed to interested programs	
			path	indicates the default path for most data files, especially output files, has changed	
	root	indicates that the path root (above path and dpath) has been set	returned to Batch_Sort returned struct has error flag (0 = OK) and perhaps some data		
	wizard-return	return after a "wizard-action" has completed			
image				main image display or clone (or 3D stack images)	
OnNotify_image	Incoming	batch-operations-open batch-rgb-open	simple way to test if Operations window open for batch operations. simple way to test if RGB window open for batch operations.	the main GeoPIXE routine	
			batch-save		Save various options for current images (e.g. correct X, save HTML, CSV tables) from 'Batch_Sort'
	corr-analyze-clear	returned from 'corr' to clear pixels inside spline			
	corr-analyze-spline	returned from 'corr' show pixels inside spline			
	dpath	set a specific path for raw data files, which can be eslewhere (e.g. Maia blog files)			
	image-analyze-all-clear	clear all markers (from Clone)			
	image-analyze-clear	clear current marker/shape on display (from Clone)			
	image-analyze-mark	plot current marker/shape (from Clone)			
	image-analyze-mode	change analyze mode (shape selection mode: "+" include, "-" exclude) (from Clone)			
	image-analyze-q	select pixels by passed 'q' vector (from Clone)			
	image-analyze-type	change analyze type (shape style: Distance, Box, Circle, ...) (from Clone)			
	image-clear-all-marks	returned from 'maia_launch' on NewRun			
	image-clone	another cloned 'image' has read in new image data to be displayed here too (but clone "owns" data)			
	image-corr-clear	returned from 'image' to clear pixels inside spline			
	image-corr-q	returned from 'clone' to pass on corr qc (pixel selection vector)			
	image-display	redisplay images, returned from 'image_select'			
	image-elements	change list of element labels.			
	image-kill-regions-all-planes	zero images within region pixels for all regions (from 'image_table')			

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			image-match-centroids	match in XANES stack regions	
			image-operations-closed	Operations window has closed.	
			image-process	execute a selected image processing function (from 'image_process')	
			image-region-clear	returned from 'image_table' before region-select	
			image-region-select	returned from 'image_table' to show a region from table	
			image-region-update	returned from 'image_table' to update concs for selected region	
			image-region-update-one	returned from 'image_table' to update concs for selected region (pnly current region row)	
			image-rgb-closed	RGB window has closed.	
			images	another app has read in new image data to be displayed here (local process will "own" data)	
			path	set the default path for most data files, especially output files	
			snapshot	was to snapshot the position and size of a window	experimental, not used now
			wizard-action	Wizard command request, or "open-test" probe of open wndow status.	These contain a "command" to be executed and arguments returned to Batch_Sort
many	Outgoing		done-save	Save options done (to Batch-Sort)	
			image-analyze-all-clear	clear current marker/shape on display (to Clones)	
			image-analyze-mode	analyze type (shape style: Distance, Box, Circle, ...) has change (to Clones)	
			image-analyze-q	select pixels by 'q' vector (to Clones)	
			image-analyze-type	analyze mode (shape selection mode: "+" include, "-" exclude) has changed (to Clones)	
			image-clone	images changed (to Clone) - passes whole 'pstate'	
			image-corr-clear	clear highlighted pixels (to Corr)	
			image-corr-q	new Association pixel selection 'qc' (to Corr)	
			image-display	redisplay images	
			image-region-throttle	send pixel 'q' vector to Throttle editing routine	
			image-results	new conc results available in region table	
			images-changed	images changed (to Clone) - passes new image data 'p' struct	
			image-show-element	passes new element to Image History Display (to shows details for element plane)	
			image-update-time	send pixel 'q' vector to TimeAmp	
			path	indicates the default path for most data files, especially output files, has changed	
			spectra	Notify Spectrum Display when analyze shape is Traverse or X,Y projection spectra	
			spectrum-display	new spectra to display (or changed colour map, so redisplay spectra too)	
			wizard-return	return after wizard-action (may contain returned/passed data too)	returned struct has error flag (0 = OK) and perhaps some data
spectrum_display				Open a spectrum display window instance	
	OnNotify_Spectrum				
		Incoming	adjust-values	old, used to adjust background	experimental, not used now
			cal-ab	Energy Calibration a,b changed, for this spectrum (returned from 'Cal')	
			cal-ab-all	Energy Calibration a,b changed, for all spectra (returned from 'Cal')	
			cal-ab-ra	returned from 'Cal' to re-assign peaks to new marked E', based on Markers	
			cal-ab-ra-ab	returned from 'Cal' to re-assign peaks to new marked E' based on A,B	
			chart-display	like "spectrum-display" for Chart spectra	from Maia Launch
			correct-image-pileup	display pileup using image products	
			cut-select	returned from 'cut_setup' to draw a cut	
			detector-select	highlight a specific detector (e.g. used in Maia Control from Rates window)	
			dpath	set a specific path for raw data files, which can be eslewhere (e.g. Maia blog files)	
			fit-display	returned from 'pixe_fit' after fit or set-up change	
			groups-display	like "spectrum-display" for Groups spectra	from Maia Launch
			image-clone	to keep a local copy of images struct (e.g. for matrix details)	
			image-region-delete	delete a matching region spectrum (when a rrgion row is deleted)	from Image Table
			image-regions	keep a local copy of the regions struct pointer array	from Image Table
			image-region-select	model DA spectrum overlay for the selected region/spectrum	from Image Table
			images	to keep a local copy of images struct (e.g. for matrix details)	
			image-spectrum-throttle	model overlay, plus model Throttled spectrum	from Image_Analyze, /throttle via Image Notify
			mark-e	mark an energy (returned from 'identify')	
			mark-element	mark element lines (returned from 'identify')	
			new-results	returned from 'pixe_fit' after valid fit (passed on)	
			path	set the default path for most data files, especially output files	
			results-select	this is passed on to PIXE Fit	
			snapshot	was to snapshot the position and size of a window	experimental, not used now
			spectra	new spectra data available	returned from parent, other spectrum, or image_table

many	Outgoing	spectrum-display	update spectrum display	name set by '(*pstate).update_notify' and changes with context (spectra, chart, group display) returned from 'spectrum_select' or parent from glImage These contain a "command" to be executed and arguments to 'cut_setup' to Identify to Identify to image windows to spectrum_select to 'Spectrum History' and Spectrum_select in Maia-Control spectra cals tied to Maia parameters returned struct has error flag (0 = OK) and perhaps some data				
		spectrum-highlight	set/clear Highlight mode for spectra display					
		spectrum-load	load a specified spectrum file					
		wizard-action	Wizard command request, or "open-test" probe of open wndow status.					
		cut-results	analyze_cut' results for X or View markers					
		identify-element	switch to element marker mode					
		identify-line	identify feature at Identify marker					
		image-display	images changed (e.g. correct images for pileup contributions)					
		path	indicates the default path for most data files, especially output files, has changed					
		select-update	a spectrum has been selected					
		spectra-changed	view into spectrum has changed					
		spectrum-fit	new selected spectrum passed tp pixe_fit					
		spectrum-get-cals	tell maia-launch to copy *pm Cals to spec					
		wizard-return	return after wizard-action (may contain returned/passed data too)					
		image_table			Open the image regions window			
many	OnNotify_Image_Table	incoming	dpath	set a specific path for raw data files, which can be eslewhere (e.g. Maia blog files)				
		Outgoing	image-region-throttle					
			image-results	new image data, which may have cleared regions				
			image-results-clear	table data already cleared in 'image', but clear table as viewed				
			path	set the default path for most data files, especially output files				
			snapshot	was to snapshot the position and size of a window	experimental, not used now			
			wizard-action	Wizard command request, or "open-test" probe of open wndow status.	These contain a "command" to be executed and arguments			
			dpath	indicates that the specific path for raw data files (e.g. Maia blog files) has changed				
			image-kill-regions-all-planes	zero images within regions, for all regions	done in 'image'			
			image-match-centroids	shift XANES images to match feature centroids	done in 'image'			
			image-region-clear	in response to an Update (used to clear shape elsewhere)				
			image-region-delete	image table row was deleted - Delete selected region and spectrum	(Spectrum Select must be open)			
			image-regions	regions have changed				
			image-region-select	table row selected				
			image-region-update	in response to an Update	region applied to image again			
			image-region-update-one	in response to an Update - single row	region applied to image again			
			image-results	table has changed				
			image-spectrum-throttle	for display of throttle overlay on spectrum				
			spectra	Regions and matching spectra have been loaded, infor Spectrum Display				
			wizard-return	return after wizard-action (may contain returned/passed data too)	returned struct has error flag (0 = OK) and perhaps some data			
			Spectrum_select			Open the spectrum selection window		
OnNotify_Select	incoming	array-select	subset of spectra selected for display by 'detector_select' pop-up					
		image-region-select	Row selected in Image Region table, so select corresponding row in spectrum list					
		select-highlight	highlighted spectrum changed, so update table and trigger spectrum display					
		select-update	update the table					
		spectra-changed	spectra data changed, so update spectrum pointer and the table					
		wizard-action	Wizard command request, or "open-test" probe of open wndow status.	These contain a "command" to be executed and arguments				
		<update_notify>	Update spectrum display. Defaults to 'spectrum_display'	Name depends on argument. See Note 2				
		select-highlight	select spectrum to highlight on spectrum display					
		spectrum-highlight	select row of spectrum table to highlight					
		wizard-return	return after wizard-action (may contain returned/passed data too)	returned struct has error flag (0 = OK) and perhaps some data				
		Cal			Open the spectrum eneregy calibration window			
		OnNotify_Cal	incoming	cal-x	Cal marker position returned from Spectrum_Display	from Spectrum_Display		
				path	set the default path for most data files, especially output files			
				many	outgoing	cal-ab	apply energy cal to currently selected spectrum	to Spectrum_Display
						cal-ab-all	apply energy calibration to ALL spectra	to Spectrum_Display

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		cal-ab-RA	"Re-assign" selected (marked) peaks to new energies, which maintains differences between spectra channel energy calibrations.		to Spectrum_Display
		cal-ab-RA-AB	"Re-assign" energy calibration A,B coefficients to new values, but maintain differences between spectra channel energy calibrations.		to Spectrum_Display
fit_setup			open the X-ray spectrum fit window		
fit_setup_event		incoming	mark-fit new-correct new-detectors new-filters new-yields path results-select spectrum-fit spectrum-view time-amp-pileup fit-setup-load wizard-action	select an element to fit (from Xray identify window) new multi-phase yield correction calculation matrix just done, so use that new detectors available, reload detectors and update list new filters available, reload filters and update list new yield calculation just done, so use that set the default path for most data files, especially output files row selected in fit results (pass to fit setup for refit) new spectra loaded, so initial background and initial peaks based on this update fit_setup copy of View markers from spectrum display new pileup E-T acceptance mask from 'time_amp' load a new PCM file Wizard command request, or "open-test" probe of open wndow status. return after wizard-action (may contain returned/passed data too)	from Spectrum_Display

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		outgoing	new-filters	new filters available, reload filters and update list
detector_setup				
	detector_setup_event			pop up to define a detector specification
		incoming	path	set the default path for most data files, especially output files
			mark-e	from 'ident' to mark a given energy
			detector-load	detector layout details loaded elsewhere
		outgoing	detector-setup-load	convey new detector layout to 'detector_layout' mimic display
			new-detectors	new detectors available, reload detectors and update lists elsewhere
corr				
	OnNotify_corr			The Association window, plotting element X versus Y for all (selected) image pixels
		incoming	corr-analyze-clear	clear elX,elY spline field
			corr-analyze-spline	sets elX,elY spline field, and subset of image pixels
			corr-display	update display
			image-analyze-all-clear	clear pixel selection
			image-analyze-q	pixel selection from image clone
			image-display	notified of a display change
			image-region-select	new region selected from image regions table (may contain elX,elY corr field)
			images-changed	flags change of image data to recalculate/display corr
			path	set the default path for most data files, especially output files
	Corr_Analyze_Spline	outgoing	corr-analyze-spline	sets elX,elY spline field, and subset of image pixels
	many		corr-analyze-clear	clear elX,elY spline field
			corr-display	update display
imageRGB				
	OnNotify_ImageRGB			element RGB image window
		incoming	path	set the default path for most data files, especially output files
			image-display	redisplay images, returned from 'image_select'
			images-changed	image data has changed
			image-analyze-mark	plot current marker/shape (from Clone)
			batch-save	Save various options for current images (e.g. correct X, save HTML, CSV tables) from 'Batch_Sort'
			wizard-action	Wizard command request, or "open-test" probe of open wndow status.
		outgoing	image-rgb-closed	window closed
			image-display	display image (w/ modified pixel selection)
			wizard-return	return after wizard-action (may contain returned/passed data too)
image_history				
	OnNotify_Image_history			display image data parameters (and stats) and history operation info for element image
		incoming	images-changed	image data has changed, update details and statistics
			image-show-element	update details, as displayed element has changed
			image-display	update details
		outgoing	none	
image_operations				
	OnNotify_Image_Process			list display of single-click image processing tools
		incoming	images-changed	image data has changed, update details and statistics
			batch-filter	request from 'batch_sort' for a filter to be applied
	OnKill_Image_Process	outgoing	image-operations-closed	window closed
			done-filter	filter done reply to 'batch_sort'
			image-process	request to 'image' to perform a selected image processing
			image-display	update image display and dependent displays (e.g. corr)
interelement_operations				
	interelement_operations_event			pop up window to perform interelment operations between element images
		incoming	images-changed	image data has changed, update interelement operation

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		outgoing	image-display	update image display and dependent displays (e.g. corr)	make_tvb' in 'image_routines' constructs the image bitmap using the selected interelement_transform and parameters, to make it visible before "apply"
pcs_clusters					
	OnNotify_pca_cluster	incoming	corr-display image-display path image-region-select images-changed image-analyze-q image-analyze-all-clear	update display notified of a display change set the default path for most data files, especially output files new region selected from image regions table (may contain elX,elY corr field) flags change of image data to recalculate/display corr pixel selection from image clone clear pixel selection	
		outgoing	corr-analyze-spline corr-analyze-clear corr-display	sets elX,elY spline field, and subset of image pixels clear elX,elY spline field update display	
correct_yield					
	correct_yield_event	Incoming	path image-region-select images-changed	set the default path for most data files, especially output files returned from 'image_table' to show a region from table flags change of image data to reselect element vectors, etc.	not used yet ...
		outgoing	image-display images new-correct	display image (w/ modified pixel selection) image data has changed and images and element vectors need to be updated in 'image' new multi-phase yield correction calculation matrix done	to 'fit_setup'
multi_image					
	multi_image_event	Incoming	image-display	notified of a display change	from 'image'
		outgoing	image-clone image-elements image-crop	another cloned 'image' has read in new image data to be displayed here too (but clone "owns" data) change list of element labels. crop of first element used to zoom others, but this Notify does not appear to be used	seems to be used in 'maia_launch' - perhaps not implemented
spectrum_history					
	OnNotify_spectrum_history	Incoming	spectra-changed	spectra data changed, so update spectrum pointer and the table	
		outgoing	none		
cuts_setup					
	cut_results_event	Incoming	path cut-results	set the default path for most data files, especially output files specification of a CUT using either X or CUT markers	from 'OnButton_Spectrum_Analyze'
		outgoing	cut-select	a CUT using either X or CUT markers sent to the spectrum for display	to 'spectrum_display'
time_amp					
	OnNotify_time_amp	Incoming	time_amp-display path image-region-select time_amp-analyze-clear images-changed image-update-time image-analyze-all-clear	update time-amp display set the default path for most data files, especially output files image region table row selected clear time-amplitude scatter plot flags change of image data to recalculate/display corr q vector from 'image' display, used to mask points displayed clear current marker/shape on display (clear q vector above)	forget how this works ...
		outgoing	time_amp-analyze-spline time_amp-analyze-clear time_amp-display path time_amp-pileup	output of spline curve details clear details update time-amp display set the default path for most data files, especially output files pileup limits available after also saved to file	Not used yet. Spline usually saved as pileup limits file Not used used in 'fit_setup' for pileup modelling